

Paragliding Intelligence Platform

Volume 20 - Master Index & Combined Architecture Package

Technical v2 - Build Reference, Cross-Volume Matrix, Dependency Map, and Package Manifest

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Purpose: This master volume turns the document suite into a navigable build system. It defines which document controls each domain, how features map to services, APIs and database ownership, and how the engineering team should execute the project without losing traceability.

1. Executive Control Summary

This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

2. Full Volume Register

This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

3. Recommended Reading Order

This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

4. Cross-Volume Dependency Map

This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

5. Feature-to-Service-to-Database Matrix

This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

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This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

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This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

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12. Package Manifest

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13. Final Engineering Backlog

This section is expanded in the master index and should be used as a navigation checkpoint for architecture, development, QA and investor review.

1. Executive Control Summary

The SkyThermal / Paragliding Intelligence Platform is not a generic weather viewer. It is a decision-intelligence system for paragliding: weather ingestion, terrain-aware scoring, local launch playbooks, AI briefings, digital twins, in-flight safety and commercial club/school/event tooling.

This Volume 20 is the control document for the full package. It prevents the architecture from becoming a set of disconnected PDFs by mapping every major feature to the controlling document, service owner, database owner, API contract and build stage.

Control Area	Primary Volume	Build Impact
Product scope and acceptance	Volume 01	Defines what must exist and what counts as done.
Technical architecture	Volume 02	Defines service boundaries, flows and non-functional constraints.
Data model	Volume 03	Controls migrations, indexes, retention and spatial/time-series storage.
API contracts	Volume 04	Controls client/server integration and backward compatibility.
Weather computation	Volume 05	Controls forecast variables, interpolation and derived weather features.
AI/ML and digital twins	Volume 06	Controls learning loops, feature store, model promotion and AI guardrails.
Mobile UX	Volume 07	Controls in-field experience, offline behavior and safety messaging.
Infrastructure	Volume 08	Controls runtime, CI/CD, monitoring, backup and cost control.
Security/privacy	Volume 09	Controls access, data protection, abuse prevention and incident response.
Testing/release gates	Volume 10	Controls validation, false-green prevention and QA gates.
Business/entitlements	Volume 11	Controls pricing, paid tiers and commercial metrics.
Legal/safety/licensing	Volume 12	Controls attribution, liability, safety wording and data-use constraints.
Local intelligence	Volume 13	Controls local playbooks, expert validation and site-specific hazards.
Hardware sensors	Volume 14	Controls stations, IoT telemetry and real-observation loops.
In-flight/rescue	Volume 15	Controls live safety, tracking, SOS and rescue operations.
Portals	Volume 16	Controls club/school/event/admin workflows.
Front end/design system	Volume 17	Controls UI components, map shell and layer registry.
Analytics/growth	Volume 18	Controls metrics, warehouse events and feedback loops.
Roadmap/sprints	Volume 19	Controls build order, epics, milestones and staffing.
Master control	Volume 20	Controls package navigation and traceability.

2. Full Volume Register

This register is the authoritative index of the v2 technical documentation set. Earlier non-technical drafts should be considered superseded by the v2 technical volumes.

Vol.	Document	Purpose	Primary Owner	Code
01	Product Requirements Document v2 Technical	Product thesis, feature catalog, personas, safety requirements, subscription scope, acceptance criteria.	Product strategy / product management	PRD
02	System Architecture Document v2 Technical	Service architecture, boundaries, pipelines,	Solution architecture	SAD

		deployment topology, integration model and engineering constraints.		
03	Database Design Document v2 Technical	PostgreSQL, PostGIS, TimescaleDB, schema ownership, indexes, retention and data quality rules.	Data architecture	DDD
04	API Specification v2 Technical	REST APIs, WebSockets, error model, rate limits, auth headers, payload contracts and service ownership.	Backend/API	API
05	Weather Engine Design v2 Technical	Weather source strategy, GRIB/NetCDF, forecast cubes, interpolation, terrain correction and paragliding variables.	Weather platform	WEA
06	AI & ML Architecture v2 Technical	Forecast correction, digital twin learning, ML feature store, LLM grounding, safety review and model promotion.	AI/ML	ML
07	Mobile & UX Specification v2 Technical	Mobile information architecture, in-field workflows, offline logic, map UX, tracking UX and interaction rules.	Mobile/product design	UX
08	Infrastructure & DevOps v2 Technical	Runtime environments, Kubernetes/K3s, CI/CD, storage, monitoring, backup and cost-control strategy.	DevOps/SRE	OPS
09	Security Architecture v2 Technical	Threat model, OIDC/OAuth2, RBAC/ABAC, API protection, privacy, audit logs and incident response.	Security	SEC
10	Testing & QA v2 Technical	Weather validation, API tests, mobile E2E, AI safety tests, security testing and release gates.	QA/Test	QA
11	Business Plan v2 Technical	Commercial strategy, tiers, pricing, SaaS metrics, sales motions, unit economics and investor milestones.	Business	BIZ
12	Legal, Safety & Data Licensing v2 Technical	Weather/source license register, attribution, GDPR, liability boundaries, AI wording controls and safety disclaimers.	Legal/compliance	LEGAL
13	Launch-Site Playbooks & Local Expert Ops v2 Technical	Local site playbooks, expert roles, moderation, hazard geometry, local rules and site approval lifecycle.	Site/community ops	PLAY
14	Hardware Weather Station & Sensor Network v2 Technical	Station hardware, LoRaWAN/LTE, MQTT ingestion, calibration, health scoring and station business model.	Hardware/IoT	IOT
15	In-Flight Safety Assistant & Rescue Mode v2 Technical	Live telemetry, in-flight safety rules, SOS sharing, rescue dashboard, search-area estimate and offline	Safety/rescue	SAFE

		safety logic.		
16	Web Admin, Club, School & Event Portal v2 Technical	Admin console, club/school workflows, event portal, moderation queues, BFF architecture and portal permissions.	Admin/organization	PORTAL
17	Design System & Front-End Engineering v2 Technical	Design tokens, component contracts, map shell, layer registry, MapLibre/deck.gl architecture and visual regression.	Frontend design system	FE
18	Data Analytics & Growth Intelligence v2 Technical	Analytics event taxonomy, forecast accuracy metrics, SaaS funnel metrics, warehouse model and privacy-aware telemetry.	Analytics/growth	ANALYTICS
19	Implementation Roadmap & Sprint Backlog v2 Technical	MVP scope, 24-week sprint plan, backlog, dependencies, staffing, release gates and acceptance criteria.	Delivery management	ROADMAP
20	Master Index & Combined Architecture Package v2 Technical	Cross-volume navigation, dependency matrix, feature-to-service map, feature-to-database map, build order and package manifest.	Program control	MASTER

3. Recommended Reading Order

Different stakeholders should read the suite in different orders. Developers should not start with the business plan; investors should not start with the database schema. The following paths prevent wasted effort.

Audience	Read First	Then Read	Why
Founder / product lead	01, 11, 19, 20	12, 13, 16, 18	Clarifies market, product scope, delivery plan and operating model.
Backend developer	02, 03, 04, 05	06, 08, 09, 10	Gives services, schema, APIs, weather engine and deployment constraints.
Frontend developer	07, 17, 04	01, 05, 09, 10	Gives map UX, components, API contracts and safety copy.
Mobile developer	07, 15, 04	09, 10, 17	Gives offline, tracking, in-flight safety and API contracts.
AI/ML engineer	06, 05, 03	10, 13, 14, 18	Gives feature pipelines, weather sources, model evaluation and feedback data.
DevOps/SRE	08, 09, 10	02, 05, 06, 20	Gives runtime, security, QA and operational dependencies.
Club/school/event operator	01, 13, 16	11, 12, 15	Gives operational workflows, legal/safety controls and commercial model.
Investor/advisor	01, 11, 19, 20	02, 06, 12, 18	Explains why the product wins and what must be built.

4. Cross-Volume Dependency Map

The platform must be built as interdependent loops, not isolated features. A weather map without a risk engine is not a paragliding product; an AI briefing without provenance is unsafe; a paid tier without entitlement enforcement leaks value.

Dependency Loop	Volumes Required	Reason
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MVP decision loop	01, 02, 03, 04, 05, 07, 09, 10	Site + weather + flyability + mobile UX + security + QA
Premium subscription loop	01, 04, 08, 09, 11, 17, 18	Entitlements + billing + UX gating + analytics
AI briefing loop	01, 04, 05, 06, 09, 10, 12	Grounded weather context + safety policy + evaluation
Digital twin loop	03, 05, 06, 13, 14, 18	Forecast samples + reality observations + ML correction + analytics
In-flight safety loop	04, 07, 09, 10, 12, 15	Mobile telemetry + alerts + privacy + safety tests
Club/school/event loop	04, 09, 11, 13, 16, 18, 19	Org roles + portals + monetization + analytics + delivery

Core runtime dependency graph:

weather-ingestion-service -> forecast-processing-service -> weather-tile-service
 forecast-processing-service -> risk-engine-service -> flyability-engine-service
 site-service + playbook-service -> risk-engine-service -> recommendation-service
 station-service + community-service -> digital-twin-service -> ml-inference-service
 tracking-service -> safety-context-service -> alert-engine -> mobile push
 billing-service -> entitlement-service -> api-gateway + frontend layer registry
 analytics-service <- all services via event gateway

5. Feature-to-Service-to-Database Matrix

This matrix is the primary traceability layer. Every major feature must have an owner service, database footprint, API contract and delivery stage. Features without an owner are not build-ready.

Feature ID	Feature	Owner Services	Database / Storage	API / Channel	Stage
F-001	Animated weather map shell	frontend-web, map-rendering-service, weather-tile-service	weather_tiles, map_layer_registry	GET /v1/map/layers, GET /tiles/{z}/{x}/{y}	MVP
F-002	Surface wind layer	weather-ingestion, forecast-processing, weather-api	forecast_site_hourly, forecast_grid_cells	GET /v1/forecast/grid, GET /v1/sites/{id}/forecast	MVP
F-003	Altitude winds 500/1000/1500/2000m	forecast-processing, vertical-interpolation-service	forecast_vertical_profiles	GET /v1/forecast/vertical-profile	MVP
F-004	Gust layer and gust-spread risk	forecast-processing, risk-engine	forecast_site_hourly, risk_scores	GET /v1/risk/site/{id}	MVP
F-005	Thermal potential	thermal-engine, ml-inference-service	derived_weather_features, thermal_predictions	GET /v1/sites/{id}/thermals	V1
F-006	Cloudbase layer	forecast-processing, cloudbase-service	derived_weather_features	GET /v1/map/layers/cloudbase	MVP
F-007	CAPE/instability layer	forecast-processing, risk-engine	derived_weather_features, risk_scores	GET /v1/map/layers/instability	V1
F-008	Rain radar and nowcast	radar-ingestion, nowcast-service	radar_frames, nowcast_hazards	GET /v1/nowcast/radar	V1
F-009	Lightning and storm alerting	lightning-ingestion, alert-engine	lightning_strikes, alert_events	GET /v1/nowcast/lightning, POST /v1/alerts/rules	V1
F-010	Visibility and low-cloud awareness	forecast-processing, risk-engine	derived_weather_features	GET /v1/sites/{id}/visibility	V1
F-011	Temperature/dewpoint layer	forecast-processing	forecast_site_hourly	GET /v1/map/layers/temp-dewpoint	MVP
F-012	Pressure trend layer	forecast-processing	forecast_site_hourly, model_runs	GET /v1/sites/{id}/pressure-trend	V1
F-013	Wave/rotor risk layer	risk-engine, terrain-	hazard_zones,	GET /v1/risk/rotor	V1

		service, digital-twin-service	risk_scores		
F-014	Foehn indicator	risk-engine, forecast-processing	derived_weather_features, risk_scores	GET /v1/risk/foehn	V1
F-015	Flyability score	flyability-engine, risk-engine, site-service	flyability_scores, risk_scores	GET /v1/sites/{id}/flyability	MVP
F-016	Drive or Don't Drive	recommendation-service, routing-adapter, alert-engine	recommendation_runs, user_favorites	GET /v1/recommendations/drive	V1
F-017	AI flight briefing	ai-briefing-service, context-builder, policy-guard	ai_briefings, briefing_context_snapshots	POST /v1/ai/briefing	MVP
F-018	Digital twin correction	digital-twin-service, feature-store, ml-inference	forecast_error_samples, correction_models	GET /v1/digital-twins/{site_id}	V2
F-019	Pilot reports and reality layer	community-service, moderation-service, station-service	pilot_observations, verified_reports	POST /v1/observations	MVP
F-020	Local expert playbooks	playbook-service, moderation-service	site_playbooks, playbook_rules, hazard_zones	GET /v1/sites/{id}/playbook	V1
F-021	Weather station network	iot-ingestion, station-service, qc-service	weather_stations, station_observations	POST /v1/iot/observations	V2
F-022	In-flight safety assistant	tracking-service, safety-context-service, alert-engine	flight_tracks, telemetry_points, safety_alerts	WS /v1/tracking/live	V2
F-023	Rescue mode	rescue-service, tracking-service, permission-service	rescue_cases, rescue_share_links	POST /v1/rescue/cases	V2
F-024	Club/school/event portals	portal-bff, org-service, event-service	organizations, memberships, events, training_sessions	GET /v1/portal/dashboard	V1
F-025	Billing/entitlements	billing-service, entitlement-service	subscriptions, entitlement_grants	GET /v1/entitlements/me	MVP

6. Service Ownership Register

A service owns a bounded domain and should own its migrations, tests, API handlers, monitoring dashboards and runbooks. Shared tables should be avoided unless explicitly documented in Volume 03.

Service	Responsibility	Owned Data	Primary Inputs	Scaling Rule
api-gateway	Public API routing, auth enforcement, rate limiting	All API traffic	JWT/OIDC claims, request metadata	Kubernetes HPA on RPS/CPU
site-service	Launch/landing/domain model	launch_sites, landing_zones, hazard_zones	PostGIS polygons/points	Cache hot sites in Redis
weather-ingestion-service	Pull forecast/radar/lightning/station data	raw_weather_objects, model_runs	Scheduled jobs, provider connectors	Scale workers by provider/run load
forecast-processing-service	Normalize forecast cubes and derive variables	forecast_grid_cells, forecast_site_hourly	GRIB/NetCDF/Zarr, canonical units	Batch workers, queue depth autoscale
weather-tile-service	Serve map raster/vector/weather layer tiles	weather_tiles, tile_cache	Tile requests, PMTiles, WebGL layers	CDN/cache first
risk-engine-service	Rotor/foehn/gust/storm safety scoring	risk_scores, derived_weather_features	Forecast + terrain + playbook rules	CPU workers, predictable latency
flyability-engine-service	GO/MAYBE/NO-GO and site score	flyability_scores	Risk + weather + pilot profile	Cache by site/hour/profile class
recommendation-service	Drive or Don't Drive, ranked sites	recommendation_runs	User location, favorites, forecast windows	Async precompute for favorites
ai-briefing-service	Grounded AI weather	ai_briefings,	Policy-checked context	Rate limit by tier, async

	summaries	briefing_context_snapshots	bundles	streaming
digital-twin-service	Site-specific error learning	forecast_error_samples, correction_models	Forecast vs reality samples	Nightly batch + online inference
ml-inference-service	Forecast correction/thermal/risk model serving	model_registry_refs, prediction_logs	Feature store vectors	CPU initially, GPU optional for CV
tracking-service	Live GPS tracking and telemetry	flight_tracks, telemetry_points	Mobile telemetry, WebSocket streams	Partition by active flight/session
alert-engine	Push/email/SMS/in-app alerts	alert_rules, alert_events	Forecast changes, safety triggers	Queue-based retry and DLQ
community-service	Pilot observations and reports	pilot_observations, comments, ratings	User content	Moderation queue scaling
portal-bff	Club/school/event/admin dashboards	Aggregates across org/service data	Role-scoped dashboard payloads	Cache org dashboards
billing-service	Stripe/app-store billing hooks	subscriptions, invoices, entitlement_grants	Webhook events	Webhook idempotency required
analytics-service	Event ingestion and product/weather metrics	analytics_events, warehouse marts	Client/server telemetry	Stream/batch ETL

7. API and Database Traceability

The implementation rule is: no UI feature should be merged unless it traces to an API endpoint, database owner, entitlement policy and QA acceptance criterion. This table groups core API namespaces by data domain.

API Namespace	Owner Service	Primary Tables	Controlling Volumes
/v1/auth/*	api-gateway, identity-provider	users, sessions, oauth_accounts	09, 04
/v1/entitlements/*	entitlement-service, billing-service	subscriptions, entitlement_grants	04, 11, 17
/v1/sites/*	site-service, playbook-service	launch_sites, landing_zones, hazard_zones, site_playbooks	03, 04, 13
/v1/forecast/*	weather-api, forecast-processing-service	model_runs, forecast_site_hourly, forecast_vertical_profiles	03, 04, 05
/v1/map/* and /tiles/*	weather-tile-service, map-rendering-service	weather_tiles, map_layer_registry, tile_cache	04, 05, 17
/v1/risk/*	risk-engine-service	risk_scores, derived_weather_features	04, 05, 10
/v1/flyability/*	flyability-engine-service	flyability_scores, recommendation_runs	01, 04, 05
/v1/ai/*	ai-briefing-service, policy-guard	ai_briefings, briefing_context_snapshots	04, 06, 12
/v1/observations/*	community-service, station-service	pilot_observations, station_observations	03, 13, 14
/v1/tracking/* and WS /live	tracking-service, safety-context-service	flight_tracks, telemetry_points, safety_alerts	04, 15
/v1/portal/*	portal-bff, org-service	organizations, memberships, portal_audit_events	16, 04, 09
/v1/analytics/*	analytics-service	analytics_events, warehouse marts	18, 09

8. Build Order and Release Gates

Release gates protect the project from building attractive UI before the forecast, safety and data foundations are reliable. The false-green risk - saying GO when conditions are unsafe - is treated as a launch blocker.

Gate	Required Evidence	Pass Condition
Gate 0 - concept quality	PRD, SAD, map-layer matrix, target countries	All MVP features trace to service/API/database owners.
Gate 1 - technical foundation	Repo, CI/CD, base API, DB migrations, auth	Developers can deploy dev/staging and run smoke tests.
Gate 2 - weather MVP	Forecast ingestion, site forecasts, map	Forecasts update reliably for selected

	layers	launch sites.
Gate 3 - decision MVP	Flyability, risk, AI briefing, alerts	No known false-green blockers in golden test set.
Gate 4 - beta launch	Mobile UX, paid tiers, legal copy, analytics	Closed beta users can plan a flying day end-to-end.
Gate 5 - public launch	Production ops, backups, SLOs, support runbooks	Incidents, payments, privacy requests and support workflows are operational.

9. MVP Implementation Slice

The MVP should be narrow but real: a small set of high-quality launch sites, weather layers, flyability scoring, AI briefing and alerts. The MVP does not need global coverage, raw ECMWF processing, rescue mode or first-party weather stations on day one.

MVP Area	Definition
Country/region scope	One launch-dense region such as Alps pilot cluster or Norway pilot cluster.
Sites	50-150 curated launch sites with safe directions, wind limits, landing info and local notes.
Weather providers	Open-Meteo + MET Norway where relevant; raw ECMWF/GFS can be V1/V2.
Map layers	Surface wind, altitude wind, gusts, cloudbase, thermals basic, CAPE/instability, rain, lightning, visibility.
Decision output	GO/MAYBE/NO-GO, flyability score, risk score, confidence, reason codes.
AI briefing	Grounded summary only; no free-form unsupported flight advice.
Alerts	Favorite-site flyability window, gust limit, rain/storm risk, model disagreement.
Paid tier	Free + Pro + Pro+ with entitlement enforcement in API and UI layer registry.
QA gate	Golden-site forecast validation and no critical false-green cases in curated test set.

10. Engineering Workstreams

These workstreams can run in parallel after the core domain model and API contracts are frozen. Each workstream must maintain its own backlog, automated tests and demo environment.

Workstream	Scope	Primary Volumes
WS-1 Product & UX	PRD, flows, design tokens, safety copy, user interviews	01, 07, 17
WS-2 Backend Platform	API gateway, auth integration, domain services, business logic	02, 04, 09
WS-3 Data & GIS	PostGIS schemas, site catalog, map data, airspace, terrain	03, 05, 13
WS-4 Weather Engine	Provider connectors, forecast processing, tiles, validation	05, 10
WS-5 AI/ML	Feature store, correction models, LLM briefing, evaluation	06, 18
WS-6 Mobile/Web Frontend	Map dashboard, layer registry, offline, tracking UX	07, 17, 15
WS-7 DevOps/Security	Runtime, CI/CD, monitoring, secrets, backup, incident response	08, 09, 10
WS-8 Commercial/Ops	Billing, portals, club/school/event ops, support	11, 16, 12

11. Quality and Safety Traceability

The platform is safety-related decision support. The QA model must trace every safety-critical output to input data, reason codes, confidence and human-readable explanation. A pretty weather map is not enough.

Safety/Quality Control	Required Rule
False-green prevention	System must not show GO when hard blockers exist: gust limit, storm/lightning, unsafe wind direction, no data, high model disagreement.
No-data policy	When required input is missing or stale, recommendation becomes UNKNOWN or MAYBE, never GO.
AI grounding	AI answers must be generated from a structured context bundle and must not invent weather, site rules or legal airspace.
Forecast validation	Every provider/model run gets freshness, completeness, spatial coverage and source attribution checks.
Entitlement enforcement	Paid layers must be enforced both in frontend and API, not only hidden in UI.
Location privacy	Live tracking and rescue sharing require explicit consent and revocable share links.
Regression suite	Golden sites and synthetic weather cases must run on every release.

12. Package Manifest

The ZIP package generated with this volume contains the v2 technical PDF/DOCX set that is present in the active workspace, plus the front-end starter ZIP where available. If a file is missing from the workspace at packaging time, it is skipped and should be regenerated from its source volume.

Vol.	Title	PDF file	DOCX file
01	Product Requirements Document v2 Technical	Paragliding_Intelligence_Platform_Volume_01_PRD_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_01_PRD_v2_Technical.docx
02	System Architecture Document v2 Technical	Paragliding_Intelligence_Platform_Technical_SAD_v2.pdf	Paragliding_Intelligence_Platform_Technical_SAD_v2.docx
03	Database Design Document v2 Technical	Paragliding_Intelligence_Platform_Volume_03_Database_Design_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_03_Database_Design_v2_Technical.docx
04	API Specification v2 Technical	Paragliding_Intelligence_Platform_Volume_04_API_Specification_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_04_API_Specification_v2_Technical.docx
05	Weather Engine Design v2 Technical	Paragliding_Intelligence_Platform_Volume_05_Weather_Engine_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_05_Weather_Engine_v2_Technical.docx
06	AI & ML Architecture v2 Technical	Paragliding_Intelligence_Platform_Volume_06_AI_ML_Architecture_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_06_AI_ML_Architecture_v2_Technical.docx
07	Mobile & UX Specification v2 Technical	Paragliding_Intelligence_Platform_Volume_07_Mobile_UX_Specification_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_07_Mobile_UX_Specification_v2_Technical.docx
08	Infrastructure & DevOps v2 Technical	Paragliding_Intelligence_Platform_Volume_08_Infrastructure_DevOps_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_08_Infrastructure_DevOps_v2_Technical.docx
09	Security Architecture v2 Technical	Paragliding_Intelligence_Platform_Volume_09_Security_Architecture_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_09_Security_Architecture_v2_Technical.docx
10	Testing & QA v2 Technical	Paragliding_Intelligence_Platform_Volume_10_Testing_QA_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_10_Testing_QA_v2_Technical.docx
11	Business Plan v2 Technical	Paragliding_Intelligence_Platform_Volume_11_Business_Plan_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_11_Business_Plan_v2_Technical.docx
12	Legal, Safety & Data Licensing v2 Technical	Paragliding_Intelligence_Platform_Volume_12_Legal_Safety_Data_Licensing_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_12_Legal_Safety_Data_Licensing_v2_Technical.docx
13	Launch-Site Playbooks & Local	Paragliding_Intelligence_Platform	Paragliding_Intelligence_Platform

	Expert Ops v2 Technical	m_Volume_13_Launch_Site_Platforms_Local_Expert_Ops_v2_Technical.pdf	m_Volume_13_Launch_Site_Platforms_Local_Expert_Ops_v2_Technical.docx
14	Hardware Weather Station & Sensor Network v2 Technical	Paragliding_Intelligence_Platform_Volume_14_Hardware_Weather_Station_Sensor_Network_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_14_Hardware_Weather_Station_Sensor_Network_v2_Technical.docx
15	In-Flight Safety Assistant & Rescue Mode v2 Technical	Paragliding_Intelligence_Platform_Volume_15_InFlight_Safety_Rescue_Mode_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_15_InFlight_Safety_Rescue_Mode_v2_Technical.docx
16	Web Admin, Club, School & Event Portal v2 Technical	Paragliding_Intelligence_Platform_Volume_16_Web_Admin_Club_School_Event_Portal_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_16_Web_Admin_Club_School_Event_Portal_v2_Technical.docx
17	Design System & Front-End Engineering v2 Technical	Paragliding_Intelligence_Platform_Volume_17_Design_System_Frontend_Engineering_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_17_Design_System_Frontend_Engineering_v2_Technical.docx
18	Data Analytics & Growth Intelligence v2 Technical	Paragliding_Intelligence_Platform_Volume_18_Data_Analytics_Growth_Intelligence_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_18_Data_Analytics_Growth_Intelligence_v2_Technical.docx
19	Implementation Roadmap & Sprint Backlog v2 Technical	Paragliding_Intelligence_Platform_Volume_19_Implementation_Roadmap_Sprint_Backlog_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_19_Implementation_Roadmap_Sprint_Backlog_v2_Technical.docx
20	Master Index & Combined Architecture Package v2 Technical	Paragliding_Intelligence_Platform_Volume_20_Master_Index_Combined_Architecture_Package_v2_Technical.pdf	Paragliding_Intelligence_Platform_Volume_20_Master_Index_Combined_Architecture_Package_v2_Technical.docx
FE	SkyThermal front-end starter layout	skythermal_frontend_layout.zip	N/A

13. Final Engineering Backlog

This backlog is not a replacement for Volume 19. It is a high-level master backlog that confirms the next implementation steps after documentation completion.

Epic ID	Epic	Scope	Owner
EPIC-WEA-01	Provider connectors	Open-Meteo/MET Norway MVP, ECMWF/GFS raw pipeline later	Weather team
EPIC-MAP-01	Map shell and layer registry	MapLibre map, layer toggles, altitude selector, timeline	Frontend team
EPIC-SITE-01	Launch-site catalog	CRUD, PostGIS geometry, launch/landing hazards, favorites	Backend/GIS
EPIC-RISK-01	Flyability and hard blockers	GO/MAYBE/NO-GO, confidence, risk reason codes	Backend/weather
EPIC-AI-01	Grounded briefing	Context builder, JSON response, policy checks, citations/provenance	AI/backend
EPIC-IOT-01	Weather station ingestion	Station identity, MQTT/webhook ingestion, QC rules	IoT/backend
EPIC-SAFE-01	In-flight safety	Tracking, offline rules, airspace/terrain alerts, SOS share	Mobile/backend
EPIC-ORG-01	Club/school/event portal	Org memberships, dashboards, task/session planning	Portal team

Minimum developer-start checklist:

- [] Create mono-repo or service repo structure
- [] Add Docker Compose dev stack: postgres/postgis, redis, minio, api, worker, web
- [] Apply initial SQL migrations from Volume 03

- [] Implement API gateway skeleton and auth middleware from Volume 04/09
- [] Implement site-service with launch CRUD and PostGIS queries
- [] Implement forecast ingestion adapter for Open-Meteo MVP
- [] Implement forecast-site-hourly materialization
- [] Implement flyability score v0 with hard blockers
- [] Implement MapLibre front-end map shell from Volume 17 / starter ZIP
- [] Implement favorite sites and selected-site right panel
- [] Implement grounded AI briefing context bundle with no hallucinated weather
- [] Add golden-site QA tests and false-green blocker suite
- [] Add staging deployment and backup/restore runbook
- [] Add billing/entitlement stubs before public beta

14. Completion Status

With Volume 20, the documentation suite is now organized enough to begin implementation. The next practical task is not another architecture PDF; it is to convert the MVP slice into a repository with migrations, OpenAPI contracts, service skeletons and the MapLibre front-end implementation.

Recommended next artifact after this volume: a developer starter repository containing Docker Compose, FastAPI service skeletons, PostgreSQL/PostGIS migrations, OpenAPI YAML, and the SkyThermal front-end shell.